Engineering a Standardized OpenEnv Environment for the Automation of Healthcare Prior Authorization and Patient Triage

The modernization of healthcare infrastructure necessitates a fundamental shift from static administrative processes to dynamic, agentic workflows capable of navigating the labyrinthine complexities of medical necessity and insurance policy compliance. At the epicenter of this challenge lies the prior authorization (PA) process, a systemic bottleneck that accounts for significant administrative waste, physician burnout, and delayed patient care within the global healthcare system.[1] While traditional automation has attempted to solve this through robotic process automation (RPA) and basic rule-based engines, these solutions often fail when confronted with the unstructured, heterogeneous nature of clinical documentation and the shifting criteria of evidence-based medical policies.[3] To bridge this gap, the OpenEnv framework offers a standardized, secure, and isolated execution environment designed specifically for the training and evaluation of reinforcement learning (RL) agents and large language model (LLM) based agents.[3] By implementing a high-fidelity "Patient Triage & Prior Authorization" environment within the OpenEnv specification, researchers can develop agents that not only parse data but exhibit the nuanced clinical reasoning required to justify expensive procedures, manage complex chronic conditions, and navigate the intricate appeals process.[1]

The OpenEnv Specification and Architectural Alignment

The development of a real-world healthcare environment begins with a strict adherence to the OpenEnv core specification, which standardizes interaction through a Gymnasium-inspired API consisting of step(), reset(), and state() methods.[5] This standardization is not merely a convenience but a requirement for production-level reinforcement learning, where consistency across diverse infrastructure is paramount.[3] The architecture leverages a containerized FastAPI server that hosts the environment logic, ensuring that the patient records and insurance policies are isolated from the agent's execution context.[2] This isolation is critical in healthcare settings where data privacy and system security are non-negotiable.[3] Furthermore, the transition from stateless HTTP to stateful WebSocket sessions (via /ws) allows for lower latency and persistent context, which is essential for multi-turn medical reasoning where an agent must maintain the history of a patient's diagnostic journey.[3]

Core API and Endpoint Requirements

To meet the rigorous demands of the OpenEnv evaluation criteria, the environment must expose specific endpoints that facilitate both automated grading and baseline reproducibility. The environment must implement a /baseline endpoint to trigger the inference script and return scores across all tasks, a /grader endpoint to provide granular performance metrics after an episode concludes, and a /tasks endpoint to enumerate the available challenges and their associated action schemas.[12] These endpoints ensure that the environment is self-documenting and compatible with automated benchmarking harnesses.[5]

The Role of Typed Pydantic Models

The OpenEnv specification mandates the use of typed Pydantic models for actions, observations, and state representations.[3] In a healthcare context, this type safety prevents common integration errors, such as malformed CPT codes or invalid ICD-10 strings, which can derail an agent's reasoning process.[3] By defining these models explicitly, the environment provides a self-documenting interface that guides the agent toward structured tool use.[2]

Environment Design: The Clinical Data Ecosystem

The "Patient Triage & Prior Authorization" (PTPA) environment is built upon two foundational data structures: the Patient Record System (PRS) and the Insurance Policy Database (IPD). The PRS simulates a longitudinal electronic health record (EHR) containing a mixture of structured data (vital signs, lab results) and unstructured clinical narratives (physician progress notes, discharge summaries, physical therapy reports).[15] The IPD, conversely, acts as the definitive source of medical necessity truth, housing the evidence-based guidelines utilized by major insurance carriers to determine coverage eligibility.[19]

Structuring the Patient Record System (PRS)

For an agent to learn effectively, the PRS must reflect the complexities of actual clinical documentation. This includes historical descriptions of chief complaints, past medical history, and co-morbid factors that might influence treatment responses.[15] The environment utilizes a synthetic data generation engine to produce realistic clinical scenarios that include common "red flags" and contraindications, which are pivotal in determining when standard policy requirements (such as a trial of physical therapy) can be bypassed.[17]

Data Category	Component Examples	Clinical Significance
Vital Signs	Height, Weight, BMI, Blood Pressure.	BMI > 35 can trigger specific GLP-1 coverage criteria.

Progress Notes	Subjective symptoms, Duration, Severity (1-10 scale).	Establishes the chronicity of pain for MRI approval.
Lab Results	HbA1c, Fasting Blood Glucose, Renal Function (eGFR).	Critical for diabetes management and exception logic.
Physical Exam	Neurological deficits, Muscle wasting, Radiculopathy.	High-weight "red flags" that justify advanced imaging.
Pharmacy History	Medication name, Dosage, Duration, Adherence.	Documents the failure of first-line therapies (Step Therapy).

The Insurance Policy Database (IPD) and Medical Necessity

The IPD serves as the environment's internal "legal" framework. It integrates real-world guidelines from organizations like Medicare (CMS) and private insurers like Aetna and Cigna.[19] These guidelines are often counter-intuitive; for instance, a CT scan is not always favored over an MRI simply due to cost; factors such as radiation exposure and diagnostic effectiveness are equally weighted.[24] The IPD stores these rules in a queryable format, requiring the agent to effectively map clinical findings in the PRS to specific policy clauses in the IPD.[4]

Task 1: Insurance Verification and Eligibility (Easy)

The initial task in the PTPA hierarchy focuses on the foundational administrative step of any medical claim: verification. The agent is provided with a patient's Member ID, a requested Procedure Code (CPT), and a diagnosis code (ICD-10). The objective is to query the IPD to determine if the insurance policy is active and if the specific procedure is listed within the "Covered Services" for that plan.[1]

Grader Logic for Verification

The grader for this easy task is programmatic and deterministic, returning a score between 0.0 and 1.0 based on the accuracy of the agent's identification of the policy status.[5]

1. **Eligibility Status (40%):** Does the agent correctly identify if the member's insurance is active?
2. **Procedure Coverage (40%):** Does the agent correctly identify if the CPT code is covered?
3. **Rationale (20%):** Does the agent cite the correct policy section in its observation?

This task serves as a baseline check for the agent's ability to use simple retrieval tools and navigate structured databases.[3]

Task 2: Medical Necessity for Advanced Imaging (Medium)

The medium-difficulty task shifts the focus toward clinical reasoning. A common administrative challenge involves the approval of Magnetic Resonance Imaging (MRI) for joint or spinal conditions.[1] Insurance policies typically require documented evidence that "conservative therapy"—such as physical therapy (PT), medication, and activity modification—has failed for a specific duration, often 4 to 6 weeks, before an MRI will be authorized.[1]

Clinical Context and Policy Contrast

The environment provides agents with varying insurance policies to simulate the fragmented nature of the U.S. healthcare market. For example, Aetna guidelines for knee MRIs might require only 3 weeks of conservative therapy (rest, ice, NSAIDs, ROM exercises), whereas other plans may strictly demand 6 weeks of documented PT.[19] The agent must retrieve the specific policy for the patient, then scan the PRS for PT notes that document the number of sessions, the dates of those sessions, and the functional outcomes.[15]

Policy Element	Requirement Detail	Source
Therapy Trial	3 to 6 weeks of PT, NSAIDs, or chiropractic care.	17
Functional Decline	Documentation that status deteriorated or failed to improve.	15
Imaging Order	Plain X-rays often required prior to MRI.	20
Urgency Bypass	"Red flags" (e.g., suspected	19

	tumor, infection, fracture).	

Grader Logic for Clinical Reasoning

The grader for Task 2 evaluates the agent's ability to synthesize temporal data and clinical outcomes.[4]

- **Evidence Extraction:** Did the agent find the exact dates of the PT sessions? [15]
- **Logical Consistency:** If the policy required 6 weeks of PT but the patient only attended for 2 weeks, did the agent correctly deny the claim? [20]
- **Exception Recognition:** If the patient presented with "True Locking" of the knee (an indicator of a torn meniscus), did the agent recognize that this justifies an immediate MRI, bypassing the PT requirement? [19]

Task 3: The Medical Exception Appeal (Hard)

The most complex task addresses the appeal process, which occurs after an initial denial. This task requires the agent to act as a specialist, searching through "Additional Evidence"—such as advanced laboratory metrics or specific diagnostic indicators—to justify a "Medical Exception" for a restricted treatment or device.[28]

The Case of Continuous Glucose Monitors (CGMs)

The hard task often centers on Diabetic Management Devices. Many health plans deny CGMs unless the patient meets specific "Step Therapy" requirements, such as multiple daily insulin injections.[26] However, exceptions are medically necessary for patients with "Problematic Hypoglycemia" or the "Dawn Phenomenon".[25]

To succeed, the agent must search the PRS for specific missing metrics:

- **Dawn Phenomenon:** Fasting blood sugars frequently exceeding 200 mg/dL.[26]
- **Hypoglycemic Unawareness:** Documented glucose readings < 54 mg/dL that the patient did not perceive.[25]
- **Glycemic Variability:** HbA1c levels that do not meet target treatments despite 3 months of adherence to current therapy.[26]

Grader Logic for Medical Exception Justification

The grader for Task 3 utilizes a multi-step verification process, often incorporating an LLM-as-a-judge to evaluate the quality of the "Letter of Medical Necessity" (LMN) generated by the agent.[14]

1. **Metric Identification (40%):** Did the agent correctly identify the "200 mg/dL" or "< 54 mg/dL" threshold in the clinical data? [25]
2. **Rationale Mapping (30%):** Did the agent link the identified metric to the specific

exception clause in the insurance policy? [26]

3. **Communication Quality (30%):** Does the appeal letter follow standard professional templates, including ICD-10 codes and physician attestations? [28]

Reward Engineering: Shaping Agentic Behavior

A meaningful reward function is critical for guiding the agent through the long-horizon reasoning required in healthcare administration. Binary end-of-episode rewards are insufficient for training agents to navigate complex medical records; instead, the PTPA environment provides "partial progress signals" that reward the agent for the successful completion of intermediate steps.[13]

Progressive Reward Structure

The reward function is shaped to encourage thorough documentation and penalize "shortcuts" or undesirable behaviors.[13]

Signal Type	Logic / Condition	Impact on Reward
Discovery Reward	Successfully retrieving the relevant insurance policy section.	+0.1
Evidence Reward	Extracting a lab value or PT note that is critical to the case.	+0.2
Logic Reward	Correctly identifying a "red flag" or exception criteria.	+0.3
Success Reward	Submitting the correct Approve/Deny/Appeal determination.	+0.4

Penalty: Loop	Repeating the same query multiple times without new results.	-0.1
Penalty: Destructive	Accessing unauthorized records or submitting false information.	-0.5

These signals ensure that the agent learns the value of each piece of clinical evidence, much like a human medical reviewer must build a case for or against a procedure.[1]

Implementation Details and OpenEnv Compliance

The PTPA environment is built using the standard OpenEnv project structure, ensuring compatibility with the broader ecosystem and ease of deployment.[37]

Action and Observation Schemas

Actions and observations are defined as Pydantic dataclasses to enforce type safety and structure.[3]

Python

```python
from pydantic import BaseModel, Field
from typing import List, Optional, Literal

class PTPAAction(BaseModel):
    action_type: Literal
    patient_id: str
    parameters: dict = Field(default_factory=dict)

class PTPAObservation(BaseModel):
    result: str
    found_evidence: Optional[List[dict]] = None
    success: bool
    error: Optional[str] = None
```

The openenv.yaml Manifest

The manifest controls the environment's configuration and deployment on the Hugging Face Hub.[37] It defines the environment's version, description, and metadata tags required for auto-discovery.[11]

YAML

```yaml
name: healthcare_prior_auth
version: "1.2.0"
description: "High-fidelity healthcare triage and prior authorization environment."
tags: ["healthcare", "legal-reasoning", "multi-step", "text-heavy"]
hardware: "cpu-basic"
endpoints:
  tasks: "/tasks"
  grader: "/grader"
  baseline: "/baseline"
```

Infrastructure and Deployment

To facilitate widespread use and reproducible research, the PTPA environment is designed for containerized execution and deployment to Hugging Face Spaces.[5]

Containerization with Docker

The environment includes a working Dockerfile that allows for one-command local setup via docker build + docker run.[7] This ensures that the environment behaves consistently across different operating systems, which is a core design principle of OpenEnv.[3]

Dockerfile

```dockerfile
# Simplified PTPA Dockerfile
FROM ghcr.io/meta-pytorch/openenv-base:latest
WORKDIR /app
COPY pyproject.toml.
RUN pip install.
COPY..
EXPOSE 8000
```

```
CMD ["uvicorn", "server.app:app", "--host", "0.0.0.0", "--port", "8000"]
```

Hugging Face Spaces Deployment

The environment is deployed as a Docker-based Space on Hugging Face, tagged with openenv for discoverability.[37] This deployment includes the built-in OpenEnv Web UI, which provides a two-pane layout for real-time interaction: human/agent dialogue on the left and state observations on the right.[5] This interface is crucial for "Human-in-the-Loop" validation of the agent's clinical justifications.[5]

Baseline Inference and Reproducibility

A core requirement of the OpenEnv specification is the provision of a baseline inference script that produces reproducible scores across all three tasks.[3] The PTPA environment utilizes an OpenAI-compatible client to run a gpt-4o-mini model against the environment's API.[36]

Reproducibility Framework

To ensure scores are reproducible, the baseline script utilizes a fixed set of evaluation seeds and records the full agent trace, including tool calls and reasoning steps.[3] This enables researchers to identify precisely where a model fails—whether due to malformed tool arguments, incorrect ordering of actions, or a fundamental failure in clinical reasoning.[4]

Task	Baseline Metric	Expected Outcome	Failure Analysis
Verification	Success Rate	0.98	Rare errors in Member ID parsing.
Necessity	Success Rate	0.42	Difficulty correlating PT session dates with duration rules.
Appeal	Success Rate	0.15	Frequent failure to find the "200 mg/dL" exception in notes.

The low baseline score for the hard task highlights the environment's value as a benchmark for

advanced reasoning models.[4]

Clinical Guidelines and Policy Reference

To maintain high fidelity, the PTPA environment logic is grounded in established medical treating guidelines (MTGs) and payer-specific clinical policies.[22] For instance, the NY State Workers' Compensation Board MTGs for maintenance care provide specific objective functional goals that must be met to support ongoing treatment.[30] These parameters—such as "maximum medical improvement" (MMI) and "permanent disability"—are encoded into the environment's truth-engine, requiring the agent to navigate professional terminology.[21]

Sample Policy Parameters for Diabetic Supplies

Parameter	Medicare/CMS Rule	Exception Condition
Testing Frequency	100 strips every 3 months.	Additional covered if insulin-treated or physician-certified.
CGM Coverage	History of problematic hypoglycemia.	Recurrent Level 2 events (< 54 mg/dL).
Special Features	Voice-output or dexterity aids.	Certified severe visual impairment or manual dexterity issues.

The environment logic requires agents to document these specific conditions, moving beyond general medical advice to precise administrative justification.[31]

Conclusion: The Strategic Value of Agentic Healthcare Automation

The "Patient Triage & Prior Authorization" environment, built on the OpenEnv specification, addresses one of the most critical inefficiencies in modern healthcare. By providing a secure, standardized, and high-fidelity sandbox, it enables the development of AI agents capable of mastering the intricate logic of medical necessity.[1] The transition from games and toys to

real-world tasks like prior authorization represents the true potential of agentic reinforcement learning. Through rigorous task hierarchy, programmatic graders, and meaningful reward shaping, this environment challenges agents to exhibit not just intelligence, but clinical precision and administrative excellence. As these agents evolve within the OpenEnv ecosystem, the goal of reducing the administrative burden on physicians while accelerating patient access to care becomes a tangible reality. The deployment of such environments to the Hugging Face Hub, supported by robust Docker-first infrastructure, ensures that this research is accessible, reproducible, and ready for production-level integration into the global healthcare infrastructure.

Works cited

1. Understanding when your insurance will approve/deny your scans - MH Imaging, accessed March 28, 2026, https://www.mhimaging.com/blogs/understanding-when-insurance-will-approve-deny-your-scans/
2. OpenEnv: Agentic Execution Environments - DeepFabric, accessed March 28, 2026, https://www.deepfabric.dev/blog/introduction_to_openenv
3. Evaluating Tool-Using Agents in Production-Oriented Environments with OpenEnv - Turing, accessed March 28, 2026, https://www.turing.com/blog/evaluating-tool-using-agents-in-production-oriented-environments-with-openenv
4. OpenEnv in Practice: Evaluating Tool-Using Agents in Real-World Environments, accessed March 28, 2026, https://huggingface.co/blog/openenv-turing
5. meta-pytorch/OpenEnv: An interface library for RL post training with environments. - GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv
6. OpenEnv: Agentic Execution Environments - Meta-PyTorch, accessed March 28, 2026, https://meta-pytorch.org/OpenEnv/index.html
7. OpenEnv/envs/README.md at main · meta-pytorch/OpenEnv · GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv/blob/main/envs/README.md
8. OpenEnv RL environments quickstart - Lightning AI, accessed March 28, 2026, https://lightning.ai/lightning-ai/studios/openenv-rl-environments-quickstart
9. OpenEnv: Run Agentic Execution Environments Locally - Sonusahani.com, accessed March 28, 2026, https://sonusahani.com/blogs/openenv
10. huggingface/openenv-course · GitHub - GitHub, accessed March 28, 2026, https://github.com/huggingface/openenv-course
11. Releases · meta-pytorch/OpenEnv - GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv/releases
12. Trace grading | OpenAI API, accessed March 28, 2026, https://developers.openai.com/api/docs/guides/trace-grading/
13. Graders | OpenAI API, accessed March 28, 2026, https://developers.openai.com/api/docs/guides/graders
14. chrisvoncsefalvay/reviewer-two-env: Reviewer Two, but it's an OpenEnv. -

GitHub, accessed March 28, 2026, https://github.com/chrisvoncsefalvay/reviewer-two-env
15. PT OT ST AT Services OON Instruction Guide - ASHLink, accessed March 28, 2026, https://www.ashlink.com/ASH/WCMGenerated/RSOONInstructGuide070317_tcm17-110384.pdf
16. Medical History Questionnaire - Integrated Sports Medicine & Physical Therapy, accessed March 28, 2026, https://www.integratedsportspt.com/files/pdf/patient-forms.pdf
17. CT/CTA/MRI/MRA PRIOR AUTHORIZATION FORM, accessed March 28, 2026, https://www.cigna.com/static/www-cigna-com/docs/ct-cta-mri-mra-prior-authorization-form.pdf
18. Medical Report Form Template - Jotform, accessed March 28, 2026, https://www.jotform.com/form-templates/medical-report-form
19. Magnetic Resonance Imaging (MRI) of the Extremities - Medical Clinical Policy Bulletins, accessed March 28, 2026, https://www.aetna.com/cpb/medical/data/100_199/0171.html
20. What are the requirements to get a lumbar spine Magnetic Resonance Imaging (MRI) approved by an insurance provider? - Dr.Oracle, accessed March 28, 2026, https://www.droracle.ai/articles/249529/what-are-the-requirements-to-get-a-lumbar-spine
21. General Criteria for MRIs - Mountain Health CO-OP, accessed March 28, 2026, https://mountainhealth.coop/general-criteria-for-mris/
22. Medical Benefit Medication Prior Authorization Criteria - Quartz Benefits, accessed March 28, 2026, https://quartzbenefits.com/wp-content/uploads/docs/providers/pharmacy/Medical-Benefit-Criteria-Library.pdf
23. Easing Prior Authorization for Advanced Imaging - AAFP, accessed March 28, 2026, https://www.aafp.org/pubs/fpm/issues/2024/0100/prior-authorization.html
24. LCD - Glucose Monitors (L33822) - CMS, accessed March 28, 2026, https://www.cms.gov/medicare-coverage-database/view/lcd.aspx?lcdid=33822
25. Therapeutic Continuous Glucose Monitors (CGM) Medical Necessity Guideline - Commonwealth Care Alliance, accessed March 28, 2026, https://www.commonwealthcarealliance.org/wp-content/uploads/2023/01/Therapeutic-Continuous-Glucose-Monitors-CGM-MNG-113_20230124.pdf
26. How to Get Ozempic and Mounjaro Covered by Cigna in New Jersey: Prior Authorization, Appeals, and Medical Necessity Requirements - Counterforce Health, accessed March 28, 2026, https://www.counterforcehealth.org/post/how-to-get-ozempic-and-mounjaro-covered-by-cigna-in-new-jersey-prior-authorization-appeals-and-medical-necessity-requirements/
27. Denied by Insurance? A Pharmacist Tells You How to Appeal - T1D Exchange, accessed March 28, 2026, https://t1dexchange.org/denied-by-insurance-a-pharmacist-tells-you-how-to-appeal/

28. WCB Medical Treatment Guidelines Frequently Asked Questions, accessed March 28, 2026, https://www.wcb.ny.gov/content/main/hcpp/MedicalTreatmentGuidelines/FAQs.jsp
29. COVERAGE AUTHORIZATION REQUESTS AND APPEALS GUIDE Composing A Letter Of Medical Necessity - Mounjaro, accessed March 28, 2026, https://mounjaro.lilly.com/assets/pdf/sample-letter-of-medical-necessity.pdf
30. INVOKAMET® Prior Authorization, Exceptions, and Appeals | J&J withMe for Healthcare Professionals, accessed March 28, 2026, https://www.jnjwithme.com/hcp/invokamet/insurance-coverage/prior-authorizations-exceptions-appeals/
31. Sample Physician Appeal Letter Please note, this is NOT a form letter and should be customized for your patient's specific sit - Breakthrough T1D, accessed March 28, 2026, https://www.breakthrought1d.org/wp-content/uploads/2024/05/Sample_Appeal_and_Exception_Letter.pdf
32. Continuous Glucose Monitoring Systems and Insulin Pumps - Mass.gov, accessed March 28, 2026, https://www.mass.gov/doc/guidelines-for-medical-necessity-determination-for-diabetes-management-devices-continuous-glucose-monitoring-systems-and-insulin-pumps/download
33. Provider Appeal Letter Sample for Prior Authorization, Medical Necessity, and Untimely Filing Denials | MD Clarity, accessed March 28, 2026, https://www.mdclarity.com/blog/provider-appeal-letter-sample
34. OpenEnv/tutorial/04-training.md at main - GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv/blob/main/tutorial/04-training.md
35. 02-deployment.md - meta-pytorch/OpenEnv - GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv/blob/main/tutorial/02-deployment.md
36. README.md - meta-pytorch/OpenEnv - GitHub, accessed March 28, 2026, https://github.com/meta-pytorch/OpenEnv/blob/main/README.md
37. README.md · burtenshaw/textarena-exp at f7600702057683f9dbcd28d5c9b31cd3dc0aae4d - Hugging Face, accessed March 28, 2026, https://huggingface.co/spaces/burtenshaw/textarena-exp/blob/f7600702057683f9dbcd28d5c9b31cd3dc0aae4d/README.md
38. Developer quickstart | OpenAI API, accessed March 28, 2026, https://developers.openai.com/api/docs/quickstart
39. GPT-4o mini Model | OpenAI API, accessed March 28, 2026, https://developers.openai.com/api/docs/models/gpt-4o-mini
40. Prior Authorization Approval Criteria - DenverHealthMedicalPlan.org, accessed March 28, 2026, https://www.denverhealthmedicalplan.org/sites/default/files/resources/document/MCD_CHP%20Prior%20Authorization%20criteria%201Q2026.pdf